Programme Name and Semester: BCA 4th Semester
Course Name (Course Code): **PHP and MySQL Lab (BCA49109)**
Class: BCA
Academic Session:2025-26

Laboratory Manual
**(PHP and MySQL Lab BCA49109)**

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**Table of Contents**

| Assignment No | Name of the Programs |
|---|---|
| 1 | a. Create a php webpage and print your name.<br>b. To write a program gets the name of the user from a form and show greeting text.<br>c. Design HTML form and retrieve the values in PHP script. |
| 2 | a. Create a php program to find odd or even number from given number<br>b. Write a php program to find maximum of three numbers<br>c. Write a PHP program to swap two numbers |
| 3 | a. Write a PHP program to compute give number multiplication table<br>b. To write a program for mathematical calculator.<br>c. Write PHP program to display current date and display using getdate() function |
| 4 | a. Write a php program to check whether given number is palindrome or not<br>b. Write a php program to check whether given number is Armstrong or not.<br>c. PHP Program to display digital clock with current time of the Server. |
| 5 | a. Write a program that displays a different message based on time of day. For example page should display “Good Morning” if it is accessed in the morning.<br>b. Write a PHP script for the following: Design a form to accept a string. Write a function to count the total number of vowels (a,e,i,o,u) from the string. Show the occurrences of each vowel from the string. |

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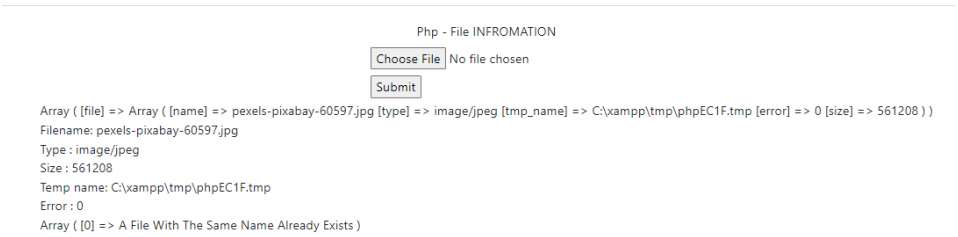
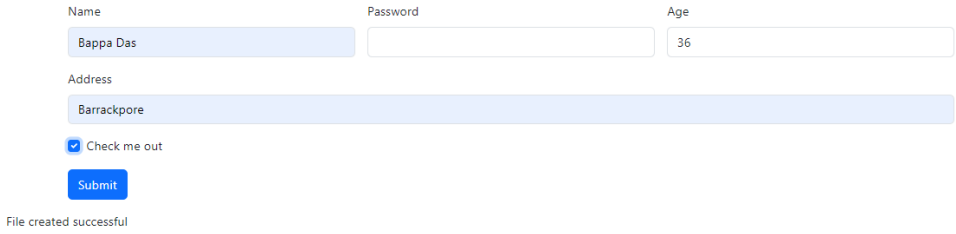
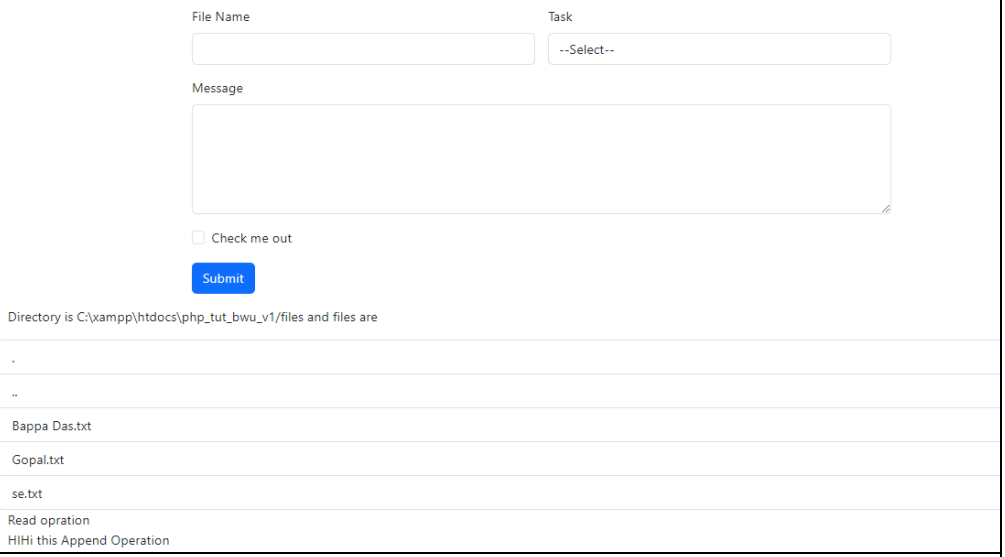
|---|---|
| | c. Write a php program to check whether given number is String palindrome or not. |
| 6 | a. Write a program to create Chess board in PHP using for loop<br>b. Write a program to calculate Electricity bill in PHP<br>Conditions:<br>● For first 50 units – Rs. 3.50/unit<br>● For next 100 units – Rs. 4.00/unit<br>● For next 100 units – Rs. 5.20/unit<br>● For units above 250 – Rs. 6.50/unit<br>You can use conditional statements.<br>c. Write a PHP Script to find out the Sum of the Individual Digits. |
| 7 | a. Write a PHP Script to find the factorial of a given number<br>b. Write a PHP Script to find the Fibonacci series of a given number.<br>c. Write an Age calculator program |
| 8 | a.Write a PHP program to create an array and store 10 names in the array. Do the following operations.<br>➢ Display the contents using for each statement.<br>➢ Display the array in a sorted order.<br>➢ Display the array without the duplicate elements<br>➢ Remove the last element and display<br>➢ Display the array in reverse order<br>➢ Insert a new element in a specified position<br>➢ Search and find out position of an element from array<br>b. Write a PHP program to create a String and store value on that string "php Program - the complete SOLUTION for WEB development". Do the following operations.<br>➢ convert lowercase string to uppercase<br>➢ convert uppercase string to lowercase<br>➢ convert first character uppercase<br>➢ convert first character of all the words uppercase<br>➢ get the last 11 characters of string<br>➢ replace the first 'the' of the following string with 'best'<br>➢ get the Second word of a sentence<br>➢ insert a string at the specified position in a given string<br>c. Write a division table program in PHP using for loop |

| | |
|---|---|
| 9 | a. Write a program in PHP get array key from given array list<br>b. Write a program in PHP to get current directory, filename and code line number<br>c. Write a program in PHP to read from directory |
| 10 | a. Write a PHP program to Get Server Information<br>b. Write a PHP program to detect remote device is mobile device or not<br>c. To create a self-processing form, use the $_SERVER['REQUEST_METHOD'] that returns the request method e.g., GET or POST.<br>➢ If the $_SERVER['REQUEST_METHOD'] is GET, you show the form.<br>➢ And if the $_SERVER['REQUEST_METHOD'] is POST, you process it |
| 11 | a. Write a PHP program to retrieve given URL information.<br>b. Write a PHP program to upload a file and get information about the upload file. |
| 12 | Write a PHP program to upload an image file within "upload" folder.<br>  <br> |
| 13 | Create a form of four field as name, address, age and password. Input name, address, age and password through form. Then create a file with name and write name, address, age, password in that file. is_file() function check the existence of file in "data" directory.<br>  <br> |
| 14 | Create a form have three input field namely File Name(text field), Message(text area),Task(list box) and a Done (submit button). A user will enter a file name and enter some text in Message (text area) then select Create Task from (list box) and submit the Done button. If file not available in data(directory) then a file will create with file name and content of Message will write in the file. if file not available in the data directory then a message will appear in Message(text area) that " File |

| | |
|---|---|
| | Doesn't exists .". when user enter a file name and select Read from list box , if file is available then content of existing file will appear in message box. Again user will enter file name and select Append from list box then file will appended with content.   |
| 15. | Design a home page in PHP, an about page in PHP and header/menu file in php, home page and about page have same header. In header/menu file two link Home, About. If click on Home link, then open home page if click about page open about page using include function in php. |

**Assignment No 1 - 4**

1. **Aim/Purpose of the Assignments:**

The aim of these PHP assignments is to provide practical knowledge of **server-side scripting using PHP** and to help students understand how dynamic web applications are developed. Through these assignments, students will learn the fundamentals of PHP programming and its integration with HTML.

The main purposes of the assignments are:

1. To understand the **basic syntax and structure of PHP programs**
2. To learn how to **embed PHP within HTML pages**
3. To work with **variables, operators, and control structures**
4. To develop skills in **form handling using GET and POST methods**
5. To perform **string, array, and mathematical operations** using PHP
6. To understand **loops, conditional statements, and functions**
7. To gain hands-on experience with **file handling and directory operations**

8. To learn **date and time functions** in PHP
9. To work with **server information and environment variables**
10. To understand **file and image upload mechanisms**
11. To develop reusable components using **include() and modular programming**

## 2. Learning Outcomes

1. Understand and apply **core PHP syntax and programming concepts**
2. Develop **dynamic web pages** by integrating PHP with HTML
3. Use **variables, operators, conditional statements, and loops** effectively
4. Handle **user input through HTML forms** using GET and POST methods
5. Perform **string manipulation and array operations** in PHP
6. Implement **mathematical and logical problem-solving** using PHP
7. Work with **date and time functions** to display server time and date
8. Create and manage **files and directories** using PHP file handling functions
9. Upload and process **files and images** securely on the server
10. Retrieve and analyze **server and URL information**
11. Design **self-processing forms** and validate user input
12. Develop **modular PHP applications** using include files
13. Build simple **utility programs** such as calculators, age calculators, and billing systems
14. Improve **logical reasoning and debugging skills**
15. Gain foundational knowledge required for **PHP-based web application development**

These learning outcomes ensure that students gain both **theoretical understanding and practical experience** in PHP programming, preparing them for advanced web development tasks.

## 3. Prerequisites:

1. Basic knowledge of computer fundamentals
2. Understanding of HTML structure and tags
3. Familiarity with CSS basics (optional but helpful)
4. Basic understanding of programming concepts such as variables, data types, and operators
5. Knowledge of control structures like if–else and loops (from any programming language)
6. Ability to use a web browser and text editor (VS Code, Notepad++, etc.)
7. Basic knowledge of web concepts such as client–server architecture
8. Installation and basic use of a local web server environment (XAMPP / WAMP / LAMP)

These prerequisites ensure that students can comfortably follow the assignments and focus on learning PHP server-side programming concepts effectively.

## 4. Software required:

To perform the PHP assignments successfully, the following software is required:

- **Operating System**: Windows / Linux / macOS

- **Web Server Package**:
  - **XAMPP / WAMP / LAMP** (includes Apache Server, PHP, and MySQL)
- **Web Browser**:
  - Google Chrome / Mozilla Firefox / Microsoft Edge
- **Text Editor / IDE**:
  - Visual Studio Code
  - Notepad++
  - Sublime Text (any one)
- **PHP Version**:
  - PHP 7.x or above
- **File Manager**:
  - To manage project folders (Windows Explorer / Linux File Manager)

These tools provide the necessary environment to write, execute, test, and debug **PHP-based web applications** effectively.

## 5. Introduction and Theory

PHP (Hypertext Preprocessor) is a widely used **server-side scripting language** designed especially for web development. It is embedded within HTML and executed on the web server, generating dynamic content that is sent to the client's web browser. PHP is open-source, platform-independent, and supports various databases, making it one of the most popular technologies for developing dynamic and interactive websites.

In PHP, scripts are executed on the **server side**, which allows developers to process user input, interact with databases, manage files, and control application logic securely. PHP supports procedural as well as object-oriented programming paradigms, enabling developers to build scalable and maintainable web applications.

The PHP programming language provides rich support for:

- **Variables and data types** to store and manipulate data
- **Operators and expressions** for logical and mathematical operations
- **Control structures** such as conditional statements and loops
- **Functions** to create reusable blocks of code
- **Form handling** using GET and POST methods
- **String and array manipulation**
- **Date and time handling**
- **File and directory management**
- **Server and environment information retrieval**

Through these assignments, students are introduced to the **core concepts of PHP programming** and its practical applications in real-world web development. The assignments focus on developing logical

thinking, understanding PHP syntax, and applying concepts to create dynamic web pages, handle user interaction, and manage server-side resources effectively.

Overall, this section builds the theoretical foundation necessary to understand how PHP works within the **client–server model**, preparing students to design and implement dynamic web applications efficiently.

## 6. Operating Procedure:

The following operating procedure should be followed to perform the PHP assignments correctly:

1. **Install Required Software**
   Install a local web server package such as **XAMPP / WAMP / LAMP** that includes Apache, PHP, and MySQL.
2. **Start the Web Server**
   Open the control panel of the server package and start the **Apache server** (and MySQL if required).
3. **Create Project Directory**
   Navigate to the server's root directory (e.g., `htdocs` in XAMPP) and create a new folder for the PHP assignments.
4. **Create PHP Files**
   Use a text editor or IDE (VS Code / Notepad++) to create PHP files with `.php` extension for each assignment program.
5. **Write the Program Code**
   Type the PHP code according to the assignment requirements, embedding PHP scripts within HTML where necessary.
6. **Save the Files**
   Save all PHP files in the created project directory inside the server root folder.
7. **Run the PHP Program**
   Open a web browser and execute the program by typing the URL:
   `http://localhost/folder_name/file_name.php`
8. **Provide Input (If Required)**
   Enter input through HTML forms where applicable and submit the form to view the output.
9. **Verify Output**
   Check the output displayed in the browser and compare it with the expected results.
10. **Debug Errors**
    If errors occur, check the code syntax, server status, and error messages. Correct the code and reload the page.
11. **File and Folder Management**
    For file-handling programs, ensure that required folders (e.g., `data`, `upload`) exist and have proper permissions.
12. **Stop Server After Use**
    After completing the assignments, stop the Apache server to free system resources.

## 7. Precautions and/or Troubleshooting

While performing the PHP assignments, the following precautions and troubleshooting steps should be considered to avoid errors and ensure smooth execution:

### Precautions

- Always save PHP files with the **`.php` extension**.
- Store all PHP files inside the **server root directory** (e.g., `htdocs` for XAMPP).
- Ensure that the **Apache server is running** before executing any PHP script.
- Use correct **syntax and proper semicolons (`;`)** in PHP statements.
- Maintain proper **indentation and code formatting** for readability.
- Validate user input to avoid unexpected errors.
- Ensure required folders (such as `data` or `upload`) **exist before file operations**.
- Set appropriate **file and folder permissions** for read/write operations.
- Avoid using deprecated PHP functions.
- Use meaningful variable names to prevent confusion.

### Troubleshooting

- If the PHP code displays as plain text, ensure PHP is installed correctly and the file is placed inside the server directory.
- Check for **syntax errors** by reviewing error messages displayed in the browser.
- Enable error reporting if output is not visible:
- `error_reporting(E_ALL);`
- `ini_set('display_errors', 1);`
- Verify form method (`GET` or `POST`) matches the PHP script.
- Check file paths carefully while performing file handling operations.
- Ensure correct **upload settings** in `php.ini` for file upload programs.
- Refresh the browser or clear cache after making code changes.
- Restart the Apache server if changes are not reflected.

Following these precautions and troubleshooting steps helps in identifying and resolving errors efficiently while working with **PHP practical programs**.

## 8. Observations

Validate the visual effects of the work done, design, check links and navigation and also validate the modifications suggested.

## 9. Calculations & Analysis

Not applicable.

## 10. Result & Interpretation

The results/output has to be shown in the computer and write it in the assignment copy.

11. **Follow-up Questions**

- What is PHP? Why is it called a server-side scripting language?
- What is the difference between client-side and server-side scripting?
- Explain how PHP is embedded within HTML.
- What are the different data types supported by PHP?
- Differentiate between GET and POST methods in PHP.
- What is the use of $_SERVER superglobal array?
- Explain the purpose of include() and require() functions.
- What is a self-processing form in PHP?
- How does PHP handle form data?
- What are conditional statements in PHP? Give examples.
- Explain different looping statements used in PHP.
- What are superglobal variables in PHP? Name any four.
- How are arrays handled in PHP? Mention types of arrays.
- What is file handling in PHP? List common file functions.
- What is the use of is_file() function?
- How do you upload a file using PHP?
- What is the importance of error handling in PHP?
- How can you display the current date and time in PHP?
- Explain the difference between echo and print.
- How can PHP be used to improve dynamic web application development?

**12. Extension and Follow-up Activities (if applicable)**

Design an interactive webpage that consists of lists, tables, hyperlinks and forms.

**13. Assessments**

As per the assessment and evaluation policy of University

**14. Suggested readings**

**Text books:** 1. PHP and MySQL Web Development, Luke Welling, Laura Thomson, Pearson,5th Edition.

**Reference Books:** 1. Programming PHP: Creating Dynamic Web Pages, Tatroe K., MacIntyre P., & Lerdorf, R., O'Reilly Press 2013, 3rd edition.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
   a) Problem definition

b) Program in corresponding language
c) Output

**Assignment No 5-7**

## 1. Aim/Purpose of the Assignments:

**Assignment 5:**
The aim of this assignment is to teach the students to apply styles to specific HTML elements using selectors, define the scope of CSS rules and determine which elements should be affected by those rules.

**Assignment 6:**
The aim of this assignment is to make students to learn the basics of box model, defining the position of elements on a web page and applying the properties such as content, padding, border, and margin.

**Assignment 7:**
The aim of this assignment is to make students to understand the concepts of positioning and types of positioning. Also insist the students to create a webpage to display the text using positioning.

## 2. Learning Outcomes

**Assignment 5:**
1. Students will learn about the basics of Cascading Style Sheets(CSS), selectors, various types of selectors and usage of selectors.
2. Helps the students to grasp the concept of selector specificity.
3. Students gains knowledge on achieving more precise targeting of elements and create complex and dynamic styles for their web pages by combining different types of selectors.
4. Selectors and avoiding overly generic ones, students can improve the performance of their web pages.

**Assignment 6:**
1. Students will learn about the how to control the layout of web pages. It helps them to style HTML elements accurately.
2. Knowledge of box model, positioning, and flexbox/grid systems enables them to create responsive and visually appealing designs that adapt to various screen sizes and devices.
3. They gain insight into selecting the right selectors and properties to manipulate fonts, colors, backgrounds, and other visual aspects to achieve the desired appearance.
4. Learning the box model helps students to create interactive websites with different components.

**Assignment 7:**
1. Students gain the ability to position HTML elements accurately and strategically on a web page.
2. They also learn about the float property, which is essential for creating multi-column layouts and organizing content.
3. Knowledge on the various positioning styles.
4. Students can position elements to adjust automatically based on the screen size, ensuring a seamless user experience across various devices.

**3. Prerequisites:**

- A foundational understanding of HTML is essential as CSS is used to style HTML elements.
- Familiarity with basic design principles and layout concepts will facilitate effective use of Cascading Style Sheets for controlling the visual presentation of web content.

**4. Software required:**

- A text editor or IDE is essential for writing and editing your CSS code.
- Web browser like Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge to preview and test your CSS-styled web pages. Browsers include developer tools that help you inspect and debug your styles.

**5. Introduction and Theory**

**Assignment 5:**

- In HTML, elements can be selected according to its id, class, type, attribute etc.
- There are several different types of selectors in CSS. The CSS selectors select HTML elements according to its id, class, type, attribute etc.
- The element selector selects the HTML element by name.
- The id selector selects the id attribute of an HTML element to select a specific element. An id is always unique within the page so it is chosen to select a single, unique element.
- It is written with the hash character (#), followed by the id of the element.
- The class selector selects HTML elements with a specific class attribute. It is used with a period character . (full stop symbol) followed by the class name.

**Assignment 6:**

- A CSS box model is a compartment that includes numerous assets, such as edge, border, padding and material. It is used to develop the design and structure of a web page. According to the CSS box model, the web browser supplies each element as a square prism.
- The CSS box model contains the different properties in CSS namely Border, Margin, Padding and Content
- Border Field is a region between the padding-box and the margin. Its proportions are determined by the width and height of the boundary.
- Margin Field consists of the area between the boundary and the edge of the border.
- Padding Field is the space around the subject area and inside the border-box. The height and the width of the padding box decide its proportions.
- The material such as text, photographs, or other digital media is included in Content Field area.
- Typically, when you assign the width and height of an attribute using the CSS width and height assets of the subject areas of that component.

**Assignment 7:**

- The position property is used in positioning an element. It can be used in conjunction with the top, right, bottom and left properties to position an element where you want it.
- There are five different position values namely, static, relative, fixed, absolute and sticky.

- The static positioning is a by default position for HTML elements. It always positions an element according to the normal flow of the page. It is not affected by the top, bottom, left and right properties.
- The fixed positioning property helps to put the text fixed on the browser. This fixed test is positioned relative to the browser window, and doesn't move even you scroll the window.
- The relative positioning property is used to set the element relative to its normal position.
- The absolute positioning is used to position an element relative to the first parent element that has a position other than static. If no such element is found, the containing block is HTML.

## 6. Operating Procedure:

- Create a HTML code with basic structure and add style and formatting information to the web pages using CSS.
- CSS can either be attached as a separate document or embedded in the HTML document itself.
- There are three methods of including CSS in an HTML document:
  - Inline styles — Using the style attribute in the HTML start tag.
  - Embedded styles — Using the <style> element in the head section of a document.
  - External style sheets — Using the <link> element, pointing to an external CSS file.
- Save and view the file. To open this file in a web browser, navigate to it, then right click on it, and select your favorite web browser.

## 7. Precautions and/or Troubleshooting

Validate the HTML tags, attributes and styles used in the HTML code.

## 8. Observations

Validate the visual effects of the work done, design, check links and navigation and also validate the modifications suggested.

## 9. Calculations & Analysis

Not applicable.

## 10. Result & Interpretation

The result or output have to be shown in computer and write it in the assignment copy

## 11. Follow-up Questions

**Assignment 5:**

- What is the difference between inline, embedded and external style sheets?
- What is the use of ruleset?
- What are the differences between adaptive design and responsive design?

**Assignment 6:**

- How to center align a div inside another div?
- How is the border-box different from the content box?
- Different Box Sizing Property?

**Assignment 7:**

- Why should we use float property in CSS?
- Explain the difference between visibility: hidden and display: none?
- What are the different ways to hide the element using CSS?

**12. Extension and Follow-up Activities (if applicable)**

Design a webpage for to display the brochure of your university events

**13. Assessments**

As per the assessment and evaluation policy of University

**14. Suggested readings**

1. Mastering HTML, CSS and Javascript Web Publishing, Laura Lemay, Rafe Colburn, Jennifer Kyrnin, BPB Publications, 2016.
2. Web Designing and Publishing, Satish Jain, M.Geetha Iyer, BPB Publications, 2020.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
    a. Problem definition
    b. Algorithm
    c. Program in corresponding language
    d. Output

Create a project implementing assignment (5-7)

Web Starter Pack
1. Lists in HTML
Ordered List
1. HTML
2. CSS
3. JavarScript
Unordered List
Notebook
Pen
Eraser
Definition List
HTML Languag for creating web pages
CSS Styles the structure
2. User Info Table
Name Age City Mobile
Aryan 21 Kolkata 9876543210
Meera 23 Delhi 9123456789
3. Navigation Pages
Go to Page 2 →
4. Registration Form
Name:
Email:
Gender: Male Female
City: Kolkata
Password:
Register

1. **Assignment No 8-10**
   **Aim/Purpose of the Assignments:**

   **Assignment 8:**
   The aim of this assignment is to learn the basics of javascript, basics of loops and arrays, declaration of variables and arrays and repetition of block using loop.

   **Assignment 9:**
   The aim of this assignment is to learn about JavaScript function concept such as declaration of function, getting value for function and passing value to the function.

   **Assignment 10:**
   The aim of this assignment is to make students to understand the concepts of validating input values, validation methods and attributes using JavaScript.

2. **Learning Outcomes**

**Assignment 8:**

1. Students gain knowledge on how variables are used to store, manipulate, and track data during program execution in JavaScript.
2. Students will learn about how to initialize an array using the constructor and populate it with a set of values.
3. Iteration of an array using a loop variable, performing an action on each element of the array.
4. Using the document.write() method to display output on the webpage.

**Assignment 9:**

- Start with HTML structure, using divs for each skill, and semantic tags for title and skill names.
- Create containers with background color, width, and border radius for each skill.
- Inside containers, define progress bars with customized styles.
- Apply specific widths using classes like “html” and “php” to represent skill levels.
- Use CSS to style HTML elements, including font size for ‘<p>’ tags and styles for containers and progress bars.

**Assignment 10:**

1. Students can understand how to handle and process user input from CSS forms, ensuring that the data submitted is accurate, complete, and meets specific criteria.
2. Students will learn about how to implement client-side validation to provide instant feedback to users about errors or incorrect input, enhancing the user experience.
3. Form validation helps prevent improper data from being submitted to the server, improving data security and integrity.

4. Students discover knowledge on how to define and enforce validation rules for various types of input, such as required fields, correct formats (e.g., email, phone numbers), and acceptable ranges.
5. Effective form validation contributes to creating user-friendly interfaces by preventing frustrating and confusing experiences due to incorrect submissions.
6. Form validation is a practical skill used in almost every web application, making it an essential part of web development.
7. Students will improve the understanding of user interactions, data handling, and front-end development.

**3. Prerequisites:**

Basic understanding of HTML, CSS.,

**4. Software required:**

- A text editor or IDE is essential for writing and editing your CSS code.
- Web browser like Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge to preview and test your CSS-styled web pages. Browsers include developer tools that help you inspect and debug your styles.

**5. Introduction and Theory**

**Assignment 8:**

- Array is a collection of data of same type, referenced by their index.
- Arrays can be created using array literals;
- Arrays can store various data types: numbers, strings, objects, functions, etc.
- Elements in an array are accessed using their index.Indices start from 0 to till [n-1]
- Arrays can be iterated using loops. For example, for, forEach etc.,

**Assignment 9:**

- Functions are fundamental building blocks of JavaScript programming and are used to organize, encapsulate, and reuse code.
- Function allows you to define a set of instructions that can be executed whenever needed.
- Function can be declared using the function keyword, followed by the function name, a list of parameters enclosed in parentheses, and the function body enclosed in curly braces.
- The function name is used to identify and call the function.
- Parameters are placeholders for values that the function expects to receive when it's called. They are listed within the parentheses of the function declaration.
- The function body contains the actual code that gets executed when the function is called. It can include any valid JavaScript statements and expressions.
- A function can optionally return a value using the return statement. The returned value can be used in the calling code for further processing.
- To execute a function, you "call" it by using its name followed by parentheses containing the required arguments or values.

**Assignment 10:**

- Validation controls in JavaScript refer to techniques and mechanisms used to ensure that user input or data adheres to certain rules or constraints before it is submitted or processed.
- These controls are commonly employed in web forms to prevent incorrect data from being sent to the server
- some common types of validation controls are Data Type Validation, Required Field Validation, Length and Size Validation, Format Validation, Range Validation, Comparison Validation, Custom Validation and Real-Time Validation.
- Data Type Validation ensures that the input matches the expected data type (e.g., numbers, strings, dates).
- Required Field Validation checks if a particular input field is filled out by the user before submitting a form. It helps prevent incomplete form submissions.
- Length and Size Validation verifies that the length of input (e.g., text, passwords) falls within specified limits.
- Format Validation validates input against a specific pattern or format, such as email addresses, phone numbers, or postal codes.
- Range Validation: verifies that numeric input falls within a certain range of values.
- Comparison Validation compares input values to each other (e.g., password confirmation).
- Custom Validation implements custom validation rules based on specific application requirements.
- Real-Time Validation provides immediate feedback to users as they enter data, often showing error messages or indicators as they type.
- JavaScript provides several techniques to implement validation controls. They are Built-in HTML Attributes, Event Listeners, Regular Expressions, Custom Functions, Third-Party Libraries.

## 6. Operating Procedure:

- Create a HTML code with basic structure and add style and formatting information to the web pages using CSS.
- Embed JavaScript code. There are typically three ways to add JavaScript to a web page:
  - Placing the JavaScript code directly inside an HTML tag using the special tag attributes such as onclick, onmouseover, onkeypress, onload, etc.
  - Embedding the JavaScript code between a pair of <script> and </script> tag.
  - Creating an external JavaScript file with the .js extension and then load it within the page through the src attribute of the <script> tag.
- Write the JavaScript for the required functionality.
- Save and view the file. To open this file in a web browser, navigate to it, then right click on it, and select your favorite web browser.

## 7. Precautions and/or Troubleshooting

Validate the HTML structure, stylesheet properties and JavaScript syntax, logic, functions used in the scripting language.

**8. Observations**

Validate the visual effects of the work done, design, data input and output, functions, check links and navigation and also validate the modifications suggested.

**9. Calculations & Analysis**

Not applicable.

**10. Result & Interpretation**

The result or output have to be shown in computer and write it in the assignment copy

**11. Follow-up Questions**

**Assignment 8:**

1. What are the different ways an HTML element can be accessed in a JavaScript code?
2. How do you empty an array in JavaScript?
3. How do you remove duplicates from a JavaScript array?

**Assignment 9:**

1. What is the use of callbacks?
2. What are the important properties of an anonymous function in JavaScript?
3. What is the difference between .call() and .apply()?

**Assignment 10:**

1. What are the pop-up boxes available in JavaScript?
2. What is JavaScript unit testing
3. How can you submit a form using JavaScript?

**12. Extension and Follow-up Activities (if applicable)**

Design a webpage for to user registration for intercollegiate meet with validation controls.

**13. Assessments**

As per the assessment and evaluation policy of University

**14. Suggested readings**

1. Mastering HTML, CSS and Javascript Web Publishing, Laura Lemay, Rafe Colburn, Jennifer Kyrnin, BPB Publications, 2016.
2. Web Designing and Publishing, Satish Jain, M.Geetha Iyer, BPB Publications, 2020.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
    a. Problem definition

b. Algorithm
c. Program in corresponding language
d. Output

1. **Assignment No 11-13**
   **Aim/Purpose of the Assignments:**

   **Assignment 11:**
   The aim of this assignment is to learn the basics of jQuery, ready function, creating animations, functions, creating smooth transition when showing or hiding content and using selectors in jQuery

   **Assignment 12:**
   The aim of this assignment is to learn about event handling using jQuery.

   **Assignment 13:**
   The aim of this assignment is to learn about jQuery UI accordion, methods of using accordion.

2. **Learning Outcomes**

**Assignment 11:**

1. Students will learn how to use jQuery's ready function and selectors to efficiently interact with and manipulate elements in the Document Object Model (DOM).
2. Understanding how to bind events, such as clicks or animations, to elements using jQuery allows you to create responsive and engaging user interfaces.
3. Learning to create animations and smooth transitions using jQuery provides you with the ability to enhance user experience by adding visually appealing effects to your web pages.
4. Studying the creation of smooth transitions for showing and hiding content is particularly useful for building UI components like slideToggle.

**Assignment 12:**

1. Students will get sound knowledge on various event handling methods such as mouse events, key events etc.
2. Knowledge of various events such as ‘click’, ‘dblclick’,’keydown’,’keypress’ etc.,
3. Students will learn about how to use jQuery's event binding methods, such as .on(), .click(), .hover(), etc., to attach event handlers to HTML elements.
4. By studying the events, students will gain the ability to make your web pages more interactive and engaging.
5. Students can enhance their knowledge in handling events for multiple elements efficiently, even if they are dynamically added or removed from the DOM.

**Assignment 13:**

1. Studying accordion sections involve adding, removing, and toggling the visibility of content based on user interactions.
2. Students will learn to design and implement accordions to create visually appealing and user-friendly interfaces.
3. Accordion elements help organize content and reduce clutter, leading to improved user comprehension and engagement.
4. Implementing smooth animations when opening and closing accordion sections contributes to a professional user experience and jQuery's animation methods like slideUp() and slideDown() to achieve these effects.

5. Working with accordions help the students to know about how to structure content in a way that is easily accessible and manageable.
6. Creating responsive accordions that adapt to different screen sizes and devices is a valuable skill for modern web development

**3. Prerequisites:**

- Basic understanding of HTML, CSS, and JavaScript concepts, as jQuery is a JavaScript library.
- Familiarity with the Document Object Model (DOM) and selecting/manipulating HTML elements will enable effective use of jQuery for creating interactive and dynamic web content.

**4. Software required:**

- A text editor or IDE is essential for writing and editing your JavaScript code, Web Browser, Coding language like Visual Studio Code, Sublime Text, Atom, JetBrains WebStorm, or Eclipse with JavaScript plugins.

**5. Introduction and Theory**

**Assignment 11:**

- The ready function in jQuery, written as $(document).ready(...), is used to specify a block of code that should be executed once the DOM (Document Object Model) of a web page is fully loaded and ready for manipulation.
- It ensures that your JavaScript code runs only after the entire HTML document has been parsed, preventing issues that may arise from interacting with elements before they are available.
- The ready function is not directly related to creating the slideUp and slideDown animation methods in jQuery. However, you can use the ready function to ensure that your animation code is executed only after the DOM is fully loaded.
- For example, when a button is clicked, the associated event handler uses the slideUp method to create a sliding animation effect on the targeted elements.

**Assignment 12:**

- Event handling in jQuery involves managing and responding to various user interactions, such as clicks, mouse movements, keyboard inputs, and more.
- jQuery provides a set of methods and functions to simplify the process of attaching event handlers and managing user actions.
- Event binding is the process of associating an event with an element and specifying a function to be executed when the event occurs. jQuery provides several methods for event binding, such as .click(), .hover(), .on(), etc.
- When an event occurs, jQuery automatically passes an event object to the event handler function. This object contains information about the event, such as the type of event, the element that triggered it, mouse coordinates, key codes, etc.
- Event delegation is a technique where you attach a single event handler to a parent element that encompasses multiple child elements. This is useful for dynamically generated elements or improving performance.

- You can stop the propagation of an event from bubbling up the DOM tree using the .stopPropagation() method on the event object.

**Assignment 13:**

- jQuery UI Accordian is an expandable and collapsible content holder that is broken into sections and probably looks like tabs.
- You can use the accordion () method in two forms:accordion (options) and accordion ("action", params) method.
- The accordion (options) method specifies that an HTML element and its contents should be treated and managed as accordion menus. The options parameter is an object that specifies the appearance and behavior of the menu involved.
- active, animate,collapsible, disabled are some of the commonly used options with accordion (options) method.
- The accordion ("action", params) method is used to perform an action on accordion elements, such as selecting/de-selecting the accordion menu. The action is specified as a string in the first argument (e.g., "disable" disables all menus).

## 6. Operating Procedure:

- Create a HTML code with basic structure and add style and formatting information to the web pages using CSS.
- Include jQuery in your document through freely available CDN (Content Delivery Network) links
- Write the jQuery script for the required functionality.
- Save and view the file. To open this file in a web browser, navigate to it, then right click on it, and select your favorite web browser.

## 7. Precautions and/or Troubleshooting

Validate the HTML structure, stylesheet properties and jQuery syntax, functions used.

## 8. Observations

Validate the visual effects of the work done, design, data input and output, check links and navigation and also validate the modifications suggested.

## 9. Calculations & Analysis

Not applicable.

## 10. Result & Interpretation

The result or output have to be shown in computer and write it in the assignment copy

## 11. Follow-up Questions

**Assignment 11:**

1. What are the jQuery functions used to provide effects?
2. What are the effects methods used in jQuery?
3. What is the use of toggle() method in JQuery?

**Assignment 12:**

1. How is jQuery different from other javascript frameworks?
2. What is jQuery click event?
3. Differentiate onload() and document.ready()?

**Assignment 13:**

1. Which event is triggered when the accordion is created?
2. How do you change the size of an accordion?
3. How do you collapse an accordion by default?

**12. Extension and Follow-up Activities (if applicable)**

Design an interactive webpage to display your university academic calendar.

**13. Assessments**

As per the assessment and evaluation policy of University

**14. Suggested readings**

1. Mastering HTML, CSS and Javascript Web Publishing, Laura Lemay, Rafe Colburn, Jennifer Kyrnin, BPB Publications, 2016.
2. Web Designing and Publishing, Satish Jain, M.Geetha Iyer, BPB Publications, 2020.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
   a) Problem definition
   b) Algorithm
   c) Program in corresponding language
   d) Output

Create a project using the assignment 8-13

**Assignment No 14 - 16**

1. **Aim/Purpose of the Assignments:**

**Assignment 14:**
The aim of this assignment is to learn the concepts of bootstrap, bootstrap grid system, and to create grid columns

**Assignment 15:**
The aim of this assignment is to learn about the bootstrap components, content method used for manipulation and properties of bootstrap components.

**Assignment 16:**

The aim of this assignment is to learn the concept of form, types of forms, adding help text and design an interactive form using bootstrap.

## 2. Learning Outcomes

**Assignment 14:**

1. Students will get sound knowledge on how to create responsive layouts that adapt to various screen
2. sizes and devices, including desktops, tablets, and smartphones.
3. Understand the concepts of responsive design, layout, and column structures.
4. With Bootstrap's predefined grid classes, students can easily create complex layouts without writing extensive CSS code.
5. The grid system helps maintain consistency in design and layout across different parts of your website.
6. Bootstrap's grid system provides a flexible and fluid layout that can be customized to fit your specific design needs.
7. Develop practical skills for creating responsive, visually appealing, and efficient web layouts.

**Assignment 15:**

1. Knowledge on bootstrap components and methods.
2. Students will get sound knowledge on creation of various bootstrap components such as button, alert.
3. Understanding the concept of how to use Bootstrap components, can help you create web applications that work consistently across different web browsers and devices.
4. Learning Bootstrap components introduces you to essential front-end development concepts, such as HTML structure, CSS classes, and sometimes JavaScript interactions.
5. Studying Bootstrap components provides you with a practical skill set that is relevant to modern web development.

**Assignment 16:**

1. Knowledge on how to create forms that adapt and look good on various screen sizes and devices, using Bootstrap's built-in responsive design features.
2. Gain insight on types of forms and creation of form layout.
3. Acquire knowledge about different form controls (e.g., text inputs, checkboxes, radio buttons, select menus) provided by Bootstrap and how to effectively integrate them into your forms.
4. Gain the ability to create uniform and visually appealing forms by leveraging Bootstrap's consistent styling and design elements.
5. Explore the customization options provided by Bootstrap to modify form elements, layouts, and styles according to your project's specific needs.

## 3. Prerequisites:

- Basic understanding of HTML and CSS for structuring and styling web content.
- Familiarity with JavaScript concepts for adding interactivity.

- Proficiency in using a text editor or integrated development environment (IDE) to write and manage code is also essential for effectively implementing Bootstrap components and styles.

4. **Software required:**
   - Text editor or an IDE to write and edit your HTML, CSS, and JavaScript code. Some popular choices include Visual Studio Code, Sublime Text, Atom, or JetBrains WebStorm.

5. **Introduction and Theory**

**Assignment 14:**

- The Bootstrap grid system is a responsive layout framework provided by the Bootstrap front-end development framework.
- It's a set of CSS classes and HTML markup that allows developers and designers to create consistent and responsive grid-based layouts for web pages and web applications.
- The grid system is designed to help create flexible and visually appealing layouts that automatically adapt to different screen sizes and devices.
- The grid system is based on a 12-column layout. Web developers can divide the available horizontal space into up to 12 equal columns, and then place content within these columns to create the desired layout.
- Bootstrap's grid system is responsive, and the columns will re-arrange depending on the screen size: On a big screen it might look better with the content organized in three columns, but on a small screen it would be better if the content items were stacked on top of each other.
- The Bootstrap grid system has four classes namely xs (for phones), sm (for tablets), md (for small laptops) and lg (for laptops and desktops)

**Assignment 15:**

- A Bootstrap component refers to a customizable, reusable element provided by the Bootstrap which makes the development faster and easier.
- Bootstrap components are designed to be responsive, visually appealing, and consistent across different devices and screen sizes.
- The components help streamline the development process by providing a standardized way to create common user interface elements, such as navigation bars, buttons, forms, modals, carousels, and more.
- some of the bootstrap components are Navbar, Button, Form Control, Card, Modal, Alert, Carousel, Pagination and Accordion

**Assignment 16:**

- A form facilitate user to enter input data such as name, email address, password etc, which can then be sent to server for processing.
- Bootstrap provides classes to create a variety of forms with varied styles, layouts and custom components.
- Bootstrap greatly simplifies the process of styling and alignment of form controls like labels, input fields, selectboxes, textareas, buttons, etc. through predefined set of classes.

- Bootstrap provides three different types of form layouts namely Vertical Form (default form layout), Horizontal Form, Inline Form.
- To create vertical form layouts simply use the predefined margin utility classes for grouping the labels, form controls, optional form text, and form validation messages.
- To create a horizontal form layout add the class .row on form groups and use the .col-*-* grid classes to specify the width of your labels and controls.
- Inline form is used to display a series of form controls, and buttons in a single horizontal row to compact the layout. You can do this easily by using the Bootstrap's grid classes.

**6. Operating Procedure:**

- Create a HTML program with basic structure and store the file with extension ".hrml".
- Include the Bootstrap CSS and JS files using their CDN links to convert HTML file to Bootstrap template
- Save and view the file. To open this file in a web browser, navigate to it, then right click on it, and select your favorite web browser.

**7. Precautions and/or Troubleshooting**

Validate the HTML structure, stylesheet properties, bootstrap controls and components.

**8. Observations**

Validate the visual effects of the work done, design, content verification, check links and navigation, data input and output, and also validate the modifications suggested.

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**9. Calculations & Analysis**

Not applicable.

**10. Result & Interpretation**

The results/output has to be shown in the computer and write it in the assignment copy.

11. **Follow-up Questions**

**Assignment 14:**

1. What are the important rules that you should follow while using Grids in Bootstrap?
2. What are the default Bootstrap text settings?
3. What is the purpose of the grid system?

**Assignment 15:**

1. How can one create an alert in Bootstrap?
2. What is a Button Group, and what is the class for a basic Button Group?
3. Is there any relationship between Bootstrap and JavaScript?

**Assignment 16:**

1. In Bootstrap, how do you make navigation elements?
2. Explain input groups in Bootstrap4.
3. How can you create a tabbed navigation menu?

**12. Extension and Follow-up Activities (if applicable)**

Design a webpage for online admission.

**13. Assessments**

As per the assessment and evaluation policy of University

**14. Suggested readings**

1. Mastering HTML, CSS and Javascript Web Publishing, Laura Lemay, Rafe Colburn, Jennifer Kyrnin, BPB Publications, 2016.
2. Web Designing and Publishing, Satish Jain, M.Geetha Iyer, BPB Publications, 2020.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
   a) Problem definition
   b) Algorithm
   c) Program in corresponding language
   d) Output

**Assignment No 17 - 19**

1. **Aim/Purpose of the Assignments:**

**Assignment 17:**
The aim of this assignment is to learn about download the theme, upload the theme, preview or activate the theme and setup.

**Assignment 18:**
The aim of this assignment is to learn and create web design with menus using word press. It also helps the students in editing and organizing menus.

**Assignment 19:**
The purpose of this assignment is to learn about content management using WordPress and to create different types of pages to publish, compress and optimize images, set the post URLs, create a table of contents on a new page.

2. **Learning Outcomes**

**Assignment 17:**

1. Knowledge on how to organize and display different types of content beyond traditional blog posts and pages.
2. Discover how to design responsive themes that adapt to different screen sizes and devices, ensuring a consistent and user-friendly experience across desktops, tablets, and smartphones.
3. Students become familiar with the structure of a WordPress theme, that helps to create custom page templates for specific layouts or content types, to control the presentation of individual pages.
4. Students will learn about how to enable theme customization options, to integrate widgetized areas and sidebars into theme.
5. Students gains multidisciplinary skill that encompasses design, front-end development, back-end development, and user experience considerations.

**Assignment 18:**

1. Knowledge on creating logical and user-friendly navigation structure for the website.
2. Students will learn about how to create different types of menus, such as primary menus, sidebar menus, and more to display subpages or related content.
3. Gain insights on various methods used to create menu using WordPress.
4. Students will get familiarity with managing menus by editing, restructuring and deleting menus.
5. Students will about learn how to customize the appearance of the menus, assign menus to specific locations.
6. Knowledge on adding custom links to the menus to link to external websites, documents, or other resources.
7. Menu creation helps in designing more user-friendly and visually appealing website with effective navigation.

**Assignment 19:**

1. Learning content management using WordPress equips you with essential skills for effectively creating, organizing, and presenting information on the web
2. Students get deep knowledge on the components and features of content management system.
3. Knowledge on creating and publishing different types of content, such as blog posts, articles, pages, and media files, using the intuitive WordPress editor and become proficient in formatting options available in the editor.
4. Studets learn about how to upload, manage, and insert images, videos, audio files, and other media into your content.
5. Understand of how to create internal and external hyperlinks within your content, improving navigation and user experience.
6. students learn to use categories and tags to organize and classify your content, making it easier for visitors to find related posts.
7. Understand on how to update menus and navigation links as you add new content, ensuring a smooth user experience.

## 3. Prerequisites:

- Basic knowledge of web development concepts such as HTML, CSS, and possibly PHP is recommended. Additionally, familiarity with content management systems and a willingness to learn how to navigate and customize the WordPress platform are beneficial prerequisites.

## 4. Software required:

- Text editor or an IDE to write and edit your HTML, CSS, and JavaScript code. Some popular choices include Visual Studio Code, Sublime Text, Atom, or JetBrains WebStorm.

## 5. Introduction and Theory

**Assignment 17:**

- In WordPress, a theme is a collection of files, templates, stylesheets, and sometimes additional functionality that determines the visual appearance and layout of a website. It's like a template that defines how your website's content is presented to visitors.
- Themes define the layout structure of various pages, such as the homepage, single post pages, archive pages, and more.
- Themes allow you to customize the design of your website without altering the underlying content or functionality.
- WordPress also allows you to create child themes. A child theme inherits the styles and functionality of a parent theme but allows you to make customizations without modifying the parent theme's files directly.
- Different templates within the theme control how different types of content are displayed.
- Theme creation consists of various stages such as Setup local environment, download and installing a stater theme, adding components, theme configuration and finally testing the theme.

**Assignment 18:**

- WordPress offers a wide variety of features for creating interactive website, including options for making and customizing navigation menus.
- A WordPress navigation menu is a list of links to different parts of a website. It's usually at the top of a page as a header. However, you can also place menus along the right or left side or at the bottom of a page as a footer.
- A menu shows the page structure that visitors see on your website's front end. It helps users locate what they're looking for.
- The method used to design menus are Site Editor, WP admin dashboard and WordPress menu widget.
- There are several approaches to managing WordPress menus. These methods include editing menus, restructuring menus and deleting menu items.

**Assignment 19:**

- Content management using WordPress involves creating, editing, organizing, and publishing various types of content, such as articles, pages, images, videos, and more.
- Content management systems can be classified into two parts: a content management application (CMA), which is more commonly known as the "back-end" of your CMS, and a content delivery application (CDA), or the "front-end" of your CMS
- Content creators can include internal links to navigate within the website, as well as external hyperlinks to direct visitors to other online resources.
- Content creators can optimize their material for search engines by adding metadata, meta descriptions, and keywords. This enhances the visibility and accessibility of the content online.
- WordPress allows for user engagement through comments on posts, fostering discussions and interactions among visitors.
- Users can customize the appearance of their content by selecting themes, applying different layouts, fonts, colors, and styles to match their branding and design preferences.
- Static pages such as the homepage, about page, contact page, etc., can be created and organized using menus. This shapes the overall structure and navigation of the website.
- Content can be further enriched by incorporating widgets and sidebars, which offer additional functionality and information such as recent posts, calendars, social media feeds, and more.

## 6. Operating Procedure:

- To work with WordPress, log in to the admin dashboard, where you can create and manage content by adding new posts or pages, customize the website's appearance using themes and widgets, and extend functionality with plugins

## 7. Precautions and/or Troubleshooting

Validating the WordPress session, plugins.

**8. Observations**

Validate the visual effects of the work done, design, content verification, check links and navigation, data input and output, and also validate the modifications suggested.

**9. Calculations & Analysis**

Not applicable.

**10. Result & Interpretation**

- The results/output has to be shown in the computer and write it in the assignment copy.

11. **Follow-up Questions**

**Assignment 17:**

1. What limitations does the WordPress website have?
2. How to create a page template?
3. State the difference between WordPress and Wix.

**Assignment 18:**

1. What is the menu limit in WordPress?
2. How to add a link to the menu that opens in a web?
3. How a description can be given to menu?

**Assignment 19:**

1. What do you mean by post and page in WordPress?
2. Where is the WordPress content stored?
3. How can comments be disabled in WordPress?

**12. Extension and Follow-up Activities (if applicable)**

Design a website for your university using WordPress.

**13. Assessments**

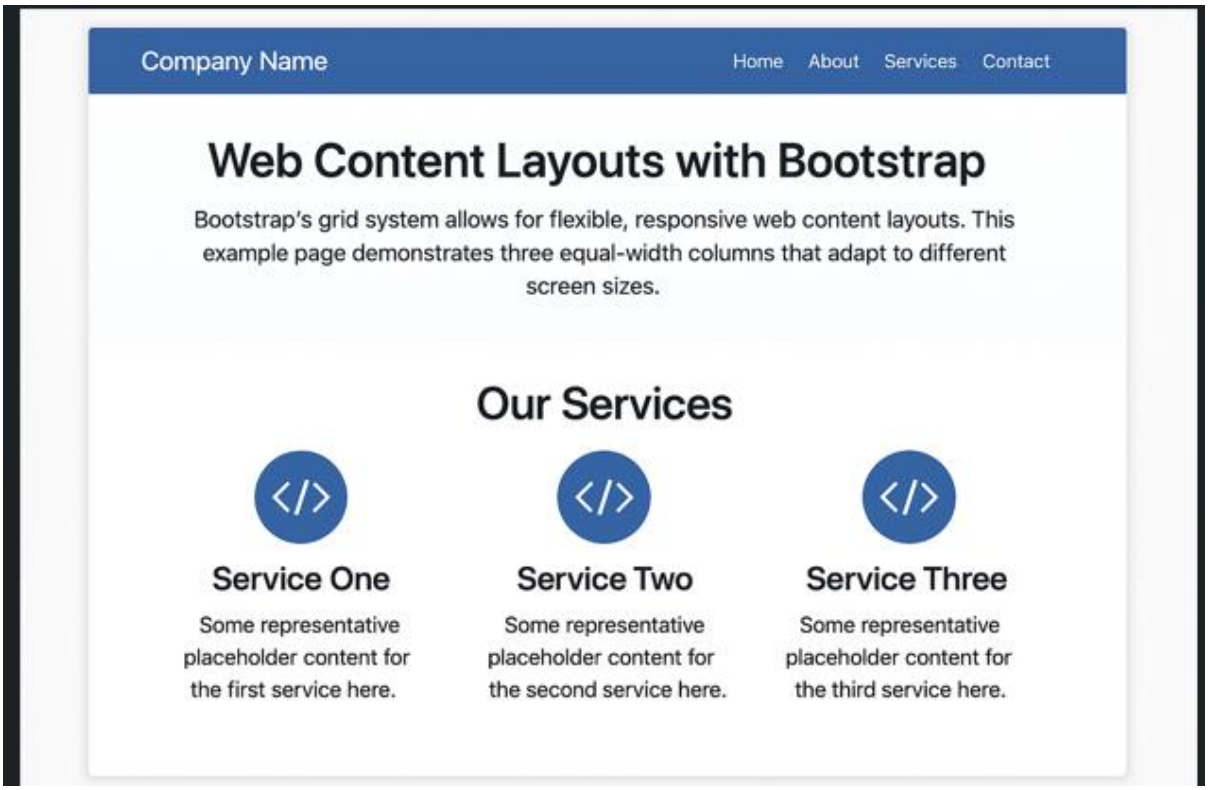
As per the assessment and evaluation policy of University

**14. Suggested readings**

1. Mastering HTML, CSS and Javascript Web Publishing, Laura Lemay, Rafe Colburn, Jennifer Kyrnin, BPB Publications, 2016.
2. Web Designing and Publishing, Satish Jain, M.Geetha Iyer, BPB Publications, 2020.

**15. Assignment copy format:**

1. Write in A4 page; No channel file will be accepted.
2. Front page and Index page format will be provided by department
3. Following points must be included while writing assignment copy
    - Problem definition
    - Algorithm
    - Program in corresponding language

- Output

Create a project using the assignments.